

Samuel Appiah Kubi

📍 Relocating to Salzburg, Austria – Sep 2026 ✉ appiahkubis14@gmail.com ☎ +233593778113

in samuel-appiah-kubi 🐙 github 🌐 Portfolio 🗣 English (C1) · German (A2, actively learning)

PROFESSIONAL SUMMARY

Geospatial AI researcher and engineer with a **First Class Honours** degree in Geomatic Engineering (KNUST, 2024), currently admitted to the **Copernicus Master's in Digital Earth** (Erasmus Mundus, Paris Lodron University of Salzburg, Sep 2026). Three years of applied research and production experience in **AI-driven multi-scale image processing, multi-modal sensor fusion, and automated target detection**, validated against real-world ground-truth measurements. My ongoing research on the **MMFF framework** (in preparation for *IEEE Transactions on Intelligent Transportation Systems*) addresses fusing data across spatial scales and sensor modalities, designing learning systems that generalise across heterogeneous surface conditions, and validating AI-derived detections against physical ground truth. I bring strong programming skills (Python, deep learning frameworks, GDAL/geospatial stack), hands-on experience with large-image tile-based inference pipelines, and a demonstrated capacity for independent, rigorous research. Driven to apply AI and multi-sensor fusion to challenging, real-world environments where ground-truth validation is difficult and generalisation across domains is critical.

RESEARCH INTERESTS

- **Multi-scale satellite and aerial image processing:** AI-driven feature detection across spatial scales, integrating broad-context imagery with high-resolution surface detail.
- **Planetary gully detection and multi-scale target prediction for Mars exploration.**
- **Multi-modal sensor fusion:** Fusing heterogeneous data streams (optical, spectral, depth, IMU, GPS) across spatial and temporal scales; fusion frameworks that generalise across surface environments.
- **Autonomous target prediction and prioritisation:** Computer vision and deep learning to identify scientifically relevant surface features and support autonomous navigation in unstructured environments.
- **Domain generalisation and algorithmic robustness:** Understanding and mitigating model failure when AI systems trained in one environment are deployed in another.
- **Validation methodology for remote sensing AI:** Designing rigorous ground-truth validation frameworks for AI-derived detections; quantifying uncertainty and confidence in automated feature identification.

RESEARCH OUTPUT

Multi-Modal Fusion Framework (MMFF) for Metric-Accurate Road Infrastructure Assessment

In preparation, 2026

Samuel Appiah Kubi et al. · Manuscript in preparation for *IEEE Transactions on Intelligent Transportation Systems*

- A five-component sensor fusion architecture fusing monocular camera (DepthAnything v2), IMU, and GPS for metric-accurate pothole dimensioning at 30 fps without LiDAR. Novel contributions: metric-scale recovery via RANSAC on ground-plane homography (GADS), event-triggered sensor selection (ETHHI), irregular-shape area estimation via Delaunay triangulation (MBTP), and temporal stability via Kalman filter (CDKF). Validated against 50 manually surveyed ground-truth measurements across 50 km of diverse road surfaces. Depth MAE 0.28 m · 62% error reduction (ETHHI) · 48.9% area-error reduction (MBTP) · 79% jitter reduction.

Smartphone-Based Road Condition Monitoring: Computer Vision and Sensor Fusion Approach

2024

Samuel Appiah Kubi · Roadessy Lab, KNUST · **2nd Place**, TRECK Annual Research Conference

- Demonstrated that vision models exploiting environmental shortcuts fail to generalise to new domains, motivating the physically informed multi-modal fusion approach developed in MMFF.

Phone Camera Calibration for Geospatial Mapping Applications

2023 – 2024

Undergraduate Thesis, KNUST · Score: 88% · **Best Final Year Project Award**

- Developed systematic lens-distortion calibration algorithms for smartphone cameras; established reproducible sensor-characterisation workflows as a prerequisite for reliable AI inference in geospatial pipelines.

OPEN-SOURCE SOFTWARE ECOSYSTEM

PyGeoVision v1.0.0 – World-Class Geospatial AI Platform

2025 – Present

github.com/appiahkubis14/PyGeoVision · [PyPI: pygeovision](https://pypi.org/project/pygeovision/) · Apache 2.0

A production-ready platform unifying satellite data acquisition (22+ providers) and geospatial AI (24 subsystems) in a single coherent API – the integration layer bridging PyGeoFetch (data acquisition) and GeoAI (inference).

- Built a production platform that unifies 22 satellite data providers and 24 geospatial AI subsystems into a single API — enabling researchers to go from satellite search to AI inference in one pipeline
- Integrated 22+ satellite data providers: USGS, Copernicus, Planet, Maxar, NASA Earthdata, Planetary Computer, AWS Earth, Sentinel Hub, Airbus OneAtlas, OpenTopography, Alaska Satellite Facility, and more
- Wrapped 24 GeoAI subsystems: semantic segmentation, object detection, classification, change detection, SAM, GroundedSAM, NASA Prithvi, DINOv3, ChangeSTAR, RF-DETR, cloud masking, super-resolution, ONNX export, canopy height estimation, and VLM captioning
- Delivered 10 end-to-end geospatial AI pipelines (building footprints, change detection, land cover, water bodies, solar detection, crop monitoring, disaster assessment, deforestation, urban growth, carbon estimation)
- Implemented 14 model architectures (UNet, SegFormer, DeepLabV3+, FCOS, RetinaNet, ViT, Siamese UNet, ChangeFormer, ESRCNN, SRCNN) and 7 automated labelers (OSM, Microsoft Building Footprints, Google Open Buildings, ESA WorldCover, Dynamic World, SAM, foundation models)
- Published to PyPI with 4 installation variants (`core`, `geoai`, `geo`, `all`); full CLI (40+ commands), YAML pipeline orchestration with cron scheduling, Docker support, and 208 passing tests

PyGeoFetch – Universal Satellite Data Pipeline

2025 – Present

github.com/appiahkubis14/pygeofetch · [PyPI: pygeofetch](https://pypi.org/project/pygeofetch/) · MIT License

Unified CLI and Python API for searching and downloading satellite imagery from 22+ global repositories with a single command; the data-acquisition backbone of PyGeoVision.

- Designed a provider-agnostic abstraction layer with authentication via system keyring (macOS/Windows/Linux), parallel downloads with adaptive concurrency, checksum verification (SHA256/MD5/SHA512), and automatic resume support for interrupted downloads
- Implemented YAML-based pipeline orchestration with cron scheduling, webhook notifications (Slack, Teams, email, custom webhooks), and 10+ post-processing actions (unzip, reproject, compress, NDVI, NDWI, composite, atmospheric correction, clip, resample, COG, merge, pan-sharpen)
- Supported 22 providers: USGS, Copernicus, NASA Earthdata, NASA Earthdata Cloud, Planet, Maxar GBDX, Airbus OneAtlas, Sentinel Hub, Google Earth Engine, OpenTopography, Alaska Satellite Facility, NOAA Big Data, Planetary Computer, JAXA ALOS, ISRO Bhuvan, INPE CBERS, ESA SciHub, DigitalGlobe Open Data, and more
- Delivered multiple output formats: STAC, GeoJSON, GeoParquet, JSON, CSV; full CQL2 filter expression support for advanced spatial and temporal querying
- Dockerized with pre-built images on Docker Hub; MIT licensed for maximum open-source compatibility

raster2pm – One-Command GeoTIFF to PMTiles Converter

2025 – Present

github.com/appiahkubis14/raster2pm · [PyPI: raster2pm](https://pypi.org/project/raster2pm/) · MIT License

Converts any GeoTIFF (orthomosaics, NDVI, NDRE, satellite imagery) to web-optimized PMTiles map tiles in a single command — eliminating complex tile pipeline setup for researchers and developers publishing geospatial data on the web.

- Built a one-command CLI (`raster2pm input.tif -o output.pmtiles`) with smart colormap auto-detection for NDVI/NDRE indices, applying scientific color ramps (RdYlGn, viridis, plasma) automatically
- Implemented memory-efficient block-by-block processing for massive rasters; tested on 64 GB orthomosaics across 36,000+ tiles — the same data scale used in the Beach Plastic Waste and Atewa pipelines
- Added built-in diagnostics (statistics, histograms, color warning detection) to validate raster integrity before conversion
- Enabled parallel processing (`--workers 8`), custom zoom levels (`--min-zoom 0 --max-zoom 18`), multiple output formats (WEBP, JPEG, PNG), and automatic stretch (`--auto-stretch`) for optimal visualization
- Published to PyPI with full documentation; output directly compatible with MapLibre, Leaflet, OpenLayers, and pmtiles.io

TECHNICAL SKILLS

- **AI and Deep Learning:** PyTorch, TensorFlow, YOLOv8/YOLOv5, U-Net, Attention U-Net, Swin-UNETR, SegFormer, CNNs, DepthAnything v2, transfer learning, domain adaptation, INT8 quantization, real-time inference optimisation, Scikit-learn
- **Multi-Scale Image Processing and Computer Vision:** Tile-based large-image inference; GDAL/OGR raster process-

ing; multi-spectral and SAR image analysis; orthomosaic generation; OBIA; OpenCV; Rasterio; Shapely; IoU-based spatial deduplication; RANSAC geometric estimation

- **Multi-Modal Sensor Fusion:** Camera + IMU + GPS fusion; Kalman filtering; RANSAC outlier rejection; ST-GNN (GAT + BiLSTM); Bayesian LSTM; T-GNN; event-triggered sensor selection; coordinate-system transformations; spatial cross-validation
- **Planetary and Remote Sensing Data:** HiRISE, CTX, CRISM, MOLA (Mars); Sentinel-1/2, Landsat, GEDI, SMAP, ERA5, IGS GIM, Swarm, DSCOVR, ACE; GNSS STAC catalogues; InSAR SBAS; Google Earth Engine (JavaScript and Python API); QGIS, ArcGIS Pro, ERDAS IMAGINE, Agisoft Metashape
- **Programming and Scientific Computing:** Python (primary: Pandas, NumPy, Matplotlib, GeoPandas, Rasterio, Shapely, OpenCV, Scikit-learn); R (tidyverse); SQL; JavaScript; GDAL scripting; Jupyter; Git
- **Infrastructure and MLOps:** Linux/Ubuntu (daily research environment); Docker; AWS (EC2, S3); Celery/Redis; CI/CD; fault-tolerant checkpoint/resume pipelines for long-running experiments

Planetary and Mars-Relevant Skills

- **Multi-scale image processing:** 64 GB orthomosaic processed across 36,000+ tiles at 9 mm GSD, directly analogous to HRSC-to-HiRISE scale transitions
- **Rare target detection:** Illegal mining sites occupying less than 4% of the landscape, transferable to gully detection on Mars where targets are sparse and morphologically variable
- **Sensor fusion:** Camera + IMU + GPS in MMFF is directly analogous to orbital and ground-based rover integration for Mars surface operations
- **Time-series and change detection:** Kalman filtering, breakpoint detection, and multi-temporal tracking of surface morphology changes
- **Planetary data familiarity:** HiRISE, CTX, CRISM, and MOLA, all used operationally in the Mars Gully Digital Twin project

KEY RESEARCH PROJECTS

Mars Gully Digital Twin · Planetary Change Detection

2025 – Present

- Developed a multi-scale planetary change detection pipeline integrating HiRISE (25 cm/px), CTX (6 m/px), CRISM spectral data, and MOLA topography to map and monitor gully morphology on Mars. An Attention U-Net trained on multi-temporal image stacks performs semantic segmentation of gully aprons, channels, and alcoves, with a Kalman filter applied to multi-epoch detections to distinguish real morphological change from instrument noise. Spatial block cross-validation on held-out HiRISE strips yields IoU greater than 0.65 and F1 greater than 0.75. Data access and versioning are managed through a STAC-compliant catalogue. Directly addresses the core objective of multi-scale target prediction: fusing orbital context (CTX) with high-resolution surface detail (HiRISE) mirrors the HRSC-to-HiRISE paradigm, and the rare-target challenge (gullies cover a small fraction of the Martian surface) parallels the illegal mining detection methodology developed in the Atewa project.

SMART-FOREST – Forest Carbon Digital Twin · REDD+ / Climate

2026 – Present

- Integrates Sentinel-1/2, GEDI LiDAR, SMAP soil moisture, and ERA5 climate reanalysis within an Object-Based Image Analysis and Temporal Graph Neural Network (T-GNN) framework to forecast above-ground biomass (AGB) over 10-year horizons under RCP climate scenarios. Spatial block cross-validation is used throughout to prevent data leakage. The multi-source data fusion strategy, combining passive optical, active microwave, and ancillary climate forcing, is directly analogous to fusing orbital spectral and topographic data streams for Mars surface characterisation.

Coastal Erosion Digital Twin · Marine and Land Change Detection

2025 – Present

- Produces 3-month shoreline change forecasts (MAE below 8 m) by fusing Sentinel-1/2, Landsat, UAV orthomosaics, and ERA5/GESLA wave climatology within a Swin-UNETR segmentation backbone and Bayesian LSTM temporal model. Digital Shoreline Analysis System (DSAS) transects provide the ground-truth change signal. The multi-temporal change detection methodology and the integration of coarse orbital imagery with high-resolution UAV data are directly transferable to detecting seasonal or inter-annual gully activity on Mars.

IonoForecaster – Space Weather Prediction · Ionospheric Scintillation

2025 – Present

- Forecasts 6-hour ionospheric scintillation (S4 index, MAE below 0.08) using a Spatio-Temporal Graph Neural Network combining Graph Attention Networks and Bidirectional LSTM, ingesting GNSS S4 observations, IGS Global Ionosphere Maps, Swarm in-situ plasma data, and DSCOVR/ACE solar wind measurements. This is the most space-physics-adjacent project in the portfolio and demonstrates familiarity with planetary space environment data streams relevant to Mars mission operations planning.

Agricultural Digital Twin · Vegetation Forecasting

2024 – Present

- An LSTM-based system forecasting 10 m NDVI time series from Sentinel-2, CHIRPS precipitation, and ESA CCI soil moisture, achieving R^2 above 0.50 and MAE below 0.07 at field parcel level. The pipeline handles missing imagery due to cloud cover via gap-filling, a challenge analogous to handling HiRISE data gaps across seasonal observation windows on Mars.

Satellite Crop Type Mapping – Foundation Model Fine-Tuning · *Earth Observation AI* 2026

- Fine-tuned the Clay Earth Observation Foundation Model on Sentinel-2 multi-temporal stacks for crop type classification at 10 m resolution, benchmarked against a ResNet-50 baseline. Spatial cross-validation across agro-ecological zones yields F1 above 0.70. This project directly informs the Mars Gully Digital Twin's approach to foundation model adaptation for sparse planetary training data.

Crop Detection and Counting from UAV Imagery – Afarinsene · *High-Resolution Object Detection* 2026

- Trained YOLOv8 on annotated Roboflow datasets from cocoa farm UAV imagery, incorporating multi-temporal analysis and uncertainty ensemble scoring to detect and count individual crop plants across smallholder and commercial farm layouts. Detection accuracy of 95.8% on the Afarinsene test set. Simultaneously diagnosed a domain generalisation failure in a companion CNN disease classifier, resolved via geospatial-aware data diversification. This work establishes the sub-metre object detection methodology that underpins the UAV component of the Mars Gully Digital Twin's ground validation strategy.

Urban Subsidence Watcher · *Edge-Deployed InSAR Monitoring* 2026

- A Tiny U-Net trained on Small Baseline Subset (SBAS) InSAR interferograms to detect urban ground deformation patterns, compressed to below 10 MB via INT8 quantization for edge deployment with inference latency below 100 ms. The model compression and edge deployment work is relevant to onboard processing constraints faced by Mars rover and lander instruments, where compute budgets are severely limited.

Atewa Illegal Mining Detector · *Rare Target Detection in Complex Terrain* 2026

- A U-Net and YOLOv8 dual-stream system fusing SAR (Sentinel-1) and optical (Sentinel-2) imagery to detect illegal artisanal mining sites (galamsey) in the Atewa Forest Reserve, Ghana. UAV flights over known sites provide auto-annotated training patches. Mining sites occupy less than 4% of the landscape, requiring hard negative mining and focal loss to prevent class imbalance. Recall of 82% and F1 of 0.79 on held-out validation tiles. The rare-target, high-false-alarm-cost detection methodology is directly transferable to gully detection on Mars, where gullies are sparse and misclassification of inactive debris aprons as active gullies has scientific consequences.

Unified Transportation Intelligence – MMFF · *IEEE ITS (In Preparation)* 2025 – 2026

- Five-component fusion architecture integrating camera (DepthAnything v2), IMU, and GPS across spatial and temporal scales: RANSAC metric-scale recovery, Delaunay triangulation for irregular feature delineation, and Kalman filter temporal stability. Validated on 50 ground-truth measurements across 50 km of heterogeneous terrain. Depth MAE 0.28 m, 79% jitter reduction, 30 fps throughput. The multi-modal sensor fusion architecture, particularly the orbital-to-rover analogy, is a core DC4 alignment point.

RESEARCH EXPERIENCE

Principal Researcher – MMFF: Multi-Modal Sensor Fusion for Infrastructure Assessment Aug 2025 – Present

TRECK / Roadessy Team, KNUST · Supervisor: Prof. Samuel Ato Andam-Akorful

- Designed MMFF as a multi-scale, multi-modal data fusion framework integrating low-resolution monocular depth with high-frequency IMU ground-truth measurements, bridging coarse-context and fine-scale observations within a single pipeline.
- Developed GADS (Geometry-Anchored Depth Scaling): recovers absolute metric scale from relative monocular depth via RANSAC regression on ground-plane homography.
- Developed ETHHI (Event-Triggered Hybrid Height Integration): selectively fuses IMU data at detected surface anomaly events, achieving 62% error reduction over monocular-only baselines.
- Developed MBTP (Minimum Bounding Triangulated Pixel): Delaunay triangulation on depth-segmented contours for accurate area estimation of geometrically irregular features – 48.9% area-error reduction.
- Validated all components against 50 manually surveyed ground-truth measurements across 50 km of heterogeneous road surfaces. Depth MAE 0.28 m; width error 6.2%; 79% jitter reduction.
- Led a three-person research team; manuscript in preparation for IEEE Transactions on Intelligent Transportation Systems.

Undergraduate Researcher – Camera Calibration and Sensor Integration 2023 – 2024

Department of Geomatic Engineering, KNUST · Supervisor: Prof. Samuel Ato Andam-Akorful

- Investigated how uncalibrated sensor characteristics propagate systematic errors through computer vision pipelines,

establishing sensor characterisation as a prerequisite for physically reliable AI inference.

- Discovered that a YOLOv8 road-condition model exploited surface-texture shortcuts rather than physically meaningful features, motivating the multi-modal fusion approach where IMU provides environment-independent measurements.
- Score: 88% – Best Final Year Project Award (highest departmental score).

APPLIED RESEARCH AND ENGINEERING EXPERIENCE

Computer Vision Research Engineer · Deep Metrics Team – Ghana

2025 – Present

- Built a real-time multi-class vehicle detection and counting system (YOLOv8 + BoT-SORT, 30 fps) with virtual line-crossing logic, class-wise statistics, and edge-optimised deployment for traffic analytics.

AI Research Engineer – Agricultural Remote Sensing and Domain Adaptation

Sep 2024 – Present

Afarinick Company Limited – Accra, Ghana

- Diagnosed catastrophic domain generalisation failure in a CNN crop-disease detector (95.8% training-domain accuracy, near-zero on smallholder farms); resolved through geospatial-aware data diversification.
- Integrated multi-modal data streams (high-resolution mobile imagery, satellite-derived NDVI/EVI indices, IoT ground sensor readings) for agricultural monitoring; managed production deployment on Linux/Docker/AWS.

Geospatial Researcher – Orbital and Ground-Based Data Fusion

2024

Rainforest Alliance – Accra, Ghana · Remote

- Developed a multi-scale hybrid monitoring system integrating Sentinel-2 (10 m, regional) with participatory plot-level ground-truth measurements; conducted automated change-detection analysis across 333.9 km².

Geospatial Data Engineer · Ghana Export-Import Bank / SKTLive LTD – Accra, Ghana

2024

- Engineered scalable ETL validation frameworks and PostGIS spatial databases; developed tile-processing pipelines converting GeoTIFF aerial imagery into multi-scale tile pyramids (zoom 5–25) via Python and GDAL.

Assistant Mine Surveyor (Intern) · Asanko Gold Ghana Ltd – Manso-Nkwanta, Ghana

Oct – Nov 2023

- Operated Total Stations, GPS/GNSS, Terrestrial Laser Scanners, and UAVs to establish ground-control points for validating satellite-derived geological models; processed dense 3D point clouds in QGIS/ArcGIS.

Pipeline Lead – Beach Plastic Waste Detection from UAV Imagery

2024

Masters Research Collaboration

- Led the end-to-end AI pipeline from raw orthomosaic (64.82 GB, 232,144 × 42,780 px, 9 mm GSD) to georeferenced detections: YOLOv8 training, tile-based inference across 36,000+ spatial tiles, overlap-aware IoU deduplication, and GeoJSON export with fault-tolerant checkpoint/resume.

EDUCATION

Paris Lodron University of Salzburg – Salzburg, Austria

Starting Sep 2026

MSc. Copernicus Master's in Digital Earth (Erasmus Mundus Joint Master Degree)

- Focus: Earth observation, geospatial data science, machine learning for satellite data, multi-scale spatial analysis

Kwame Nkrumah University of Science and Technology (KNUST) – Kumasi, Ghana

Jan 2020 – Aug 2024

BSc. Geomatic Engineering – First Class Honours (CGPA 71.74%, top departmental graduate)

- Best Final Year Project Award (88%) for sensor-fusion and computer-vision research
- Coursework: Remote Sensing, GIS and Spatial Analysis, Satellite Image Processing, Photogrammetry, Geodesy, Computer Programming, Engineering Survey, Cartography, Mathematics

HONOURS AND AWARDS

- **Admitted – Copernicus Master’s in Digital Earth** (Erasmus Mundus Joint Master Degree) · Paris Lodron University of Salzburg, 2026
- **Best Final Year Project – Highest Departmental Score (88%)** · Dept. of Geomatic Engineering, KNUST, 2024
- **First Class Honours in Geomatic Engineering** · KNUST, 2024
- **2nd Place – TRECK Annual Research Conference** · KNUST, 2024 – sensor-fusion road-condition research
- **Ghana Institution of Surveyors Students Society (GHIS-SS) Award** – academic excellence and professional service

CERTIFICATIONS

- **Machine Learning and Advanced AI Techniques** · Professional Training (TensorFlow, PyTorch, Scikit-learn)
- **Artificial Intelligence and Machine Learning** · AI/ML Certification
- **Geospatial Analytics Essentials** · Remote Sensing and Spatial Analysis
- **Remote Sensing Basics** · Satellite Image Processing and Interpretation
- **AWS Certified Solutions Architect – Professional** · Amazon Web Services
- **AWS Certified Cloud Practitioner** · Amazon Web Services

REFERENCES

- **Prof. Samuel Ato Andam-Akorful** – Associate Professor and Research Supervisor, Dept. of Geomatic Engineering, KNUST
saandam-akorful.coe@knust.edu.gh · +233 24 783 9150
Supervised undergraduate thesis and MMFF research; can speak to research capacity and scientific rigour
- **Dr. Titus Tienaah** – Lecturer, Dept. of Geomatic Engineering, KNUST and Lead, Deep Metrics Team
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Supervised computer vision research and real-time vehicle detection projects; can speak to applied AI engineering and edge deployment experience